Holographic Reduced Representations

Plate's system, step by step

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Roadmap

I	Plate's ingredients	atoms; binding by convolution; frames
II	Taking a binding apart	the involution; correlation; the noise; unitary vectors
III	Frames at work	querying; cleanup; order and similarity
IV	How much fits	counting noise; capacity; nesting; sequences
V	Fourier, from scratch	waves; the convolution theorem; the FFT; what the spectrum says about decoding
VI	Where HRR came from	holographic memory; HRR among its neighbors; what comes next
VII	Close	summary; exercises; reading

Session 11c built circular convolution as the fold of the outer product, and proved that it cannot keep argument order. Today we build the whole system Plate (1995) put on top of it, computing by hand where the vectors are small and measuring where they are not.

Where We Are

Binding has one job: hold a role and its filler in one vector, so that which-is-which survives. Session 11c's route, in four steps:

outer product	$\mathbf{r}\mathbf{f}^\top$ keeps everything, at a cost of n^2 numbers per pair
the fold	sum it along wrapped diagonals, back down to n numbers
the result	circular convolution, $(\mathbf{r} \circledast \mathbf{f})_j = \sum_k r_k \, f_{(j-k) \bmod n}$ (11c's output k is written j today)
the price	with $\mathbf{u} = (1, 2, 0, 1)$, $\mathbf{v} = (2, 0, 1, 1)$: $\mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5) = \mathbf{v} \circledast \mathbf{u}$. The fold erases the antisymmetric half, so the binding cannot tell role from filler

11c ended with three questions it left open:

1. Why is the noise small? 11c's unbinding returned 0.599, against distractors near 0.
2. How do we get from a noisy vector back to a symbol?
3. How much fits in one vector?

Plate's Holographic Reduced Representations (HRR) answer all three. That is today.

Three Questions for Today

1. **How do you take a binding apart?** Given the vector for agt $\circledast$ mark and the role agt, recover mark. Convolution has no obvious "division," so what plays its part?
2. **Why is the answer only approximately right, and how wrong is it?** We will find the error exactly at $n = 4$, then count it at $n = 1{,}024$.
3. **How much fits?** A sentence like *Hunger caused Mark to eat the fish* is several bindings inside one vector, one of them a whole clause. When do the answers stop coming back?

And a fourth, for the end: why Plate called these representations *holographic*.

Part I
Plate's Ingredients

Atoms: Random Vectors of Length About One

Every symbol gets an **atom**: each word (mark, fish), each role (agt, obj), each predicate (eat). Plate's rule: draw the n coordinates independently, with mean 0 and variance $1/n$. A normal distribution works, and so do the values $\pm 1/\sqrt{n}$.

n	**length of an atom**	**dot product of two different atoms**
64	1.00 ± 0.08	0.00 ± 0.13
1,024	1.00 ± 0.02	0.00 ± 0.03
16,384	1.00 ± 0.006	0.00 ± 0.008

Measured over 500 draws; "$\pm$" is one standard deviation.

Two facts follow. Every atom has length about 1, so a dot product between atoms is already a cosine. Two different atoms are nearly orthogonal, within about $1/\sqrt{n}$: the fact Session 11c used to explain why the fold's collisions are survivable.

Nothing in an atom's coordinates says whether it is a word or a role; that lives in how the atoms are used. We write n for the dimension, as Plate does; Session 11c's tables and the next session write d.

Binding Is Circular Convolution

index	0	1	2	3
u	1	2	0	1
v	2	0	1	1

To bind $\mathbf{u}$ and $\mathbf{v}$, output j adds every product $u_k v_m$ whose indices sum to j, wrapping around mod n:

$$(\mathbf{u} \circledast \mathbf{v})_j \;=\; \sum_k u_k \, v_{(j-k) \bmod n}$$

Session 11c: $(\mathbf{u} \circledast \mathbf{v})_k \;=\; \sum_i u_i \, v_{(k-i) \bmod n}$

Same formula, letters renamed. In 11c the output was k, the fold's group of cells (i, j) with $i + j \equiv k$. Today the output is j and k runs along $\mathbf{u}$, because Part V needs k for frequencies: 11c's k and i are today's j and k.

output	**the four products**	
$j = 0$	$u_0v_0 + u_1v_3 + u_2v_2 + u_3v_1 = 1{\cdot}2 + 2{\cdot}1 + 0{\cdot}1 + 1{\cdot}0$	4
$j = 1$	$u_0v_1 + u_1v_0 + u_2v_3 + u_3v_2 = 1{\cdot}0 + 2{\cdot}2 + 0{\cdot}1 + 1{\cdot}1$	5
$j = 2$	$u_0v_2 + u_1v_1 + u_2v_0 + u_3v_3 = 1{\cdot}1 + 2{\cdot}0 + 0{\cdot}2 + 1{\cdot}1$	2
$j = 3$	$u_0v_3 + u_1v_2 + u_2v_1 + u_3v_0 = 1{\cdot}1 + 2{\cdot}1 + 0{\cdot}0 + 1{\cdot}2$	5

So $\mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5)$: n numbers, not n^2. In each row the indices add to j, so swapping $\mathbf{u}$ and $\mathbf{v}$ only reorders the products: $\mathbf{u} \circledast \mathbf{v} = \mathbf{v} \circledast \mathbf{u}$. It is also associative, so $\mathbf{a} \circledast \mathbf{b} \circledast \mathbf{c}$ needs no brackets.

What a Binding Keeps, and What It Loses

Now with real atoms. Bind a role $\mathbf{c}$ to a filler $\mathbf{x}$ at $n = 1{,}024$ and measure:

property	**measured**	**what it means**
size	n numbers	same kind of object as an atom
unlike its inputs	$\cos(\mathbf{c} \circledast \mathbf{x}, \mathbf{c})$ and $\cos(\mathbf{c} \circledast \mathbf{x}, \mathbf{x})$: 0.00 ± 0.05	a new item
keeps similarity	$\cos(\mathbf{c} \circledast \mathbf{x}, \mathbf{c} \circledast \mathbf{y}) - \cos(\mathbf{x}, \mathbf{y})$: 0.00 ± 0.02	similar fillers, similar pairs
commutative	$\mathbf{c} \circledast \mathbf{x} = \mathbf{x} \circledast \mathbf{c}$, exactly	role and filler look alike

300 draws. For the similarity row, $\mathbf{y}$ was built with $\cos(\mathbf{x}, \mathbf{y}) = 0.6$.

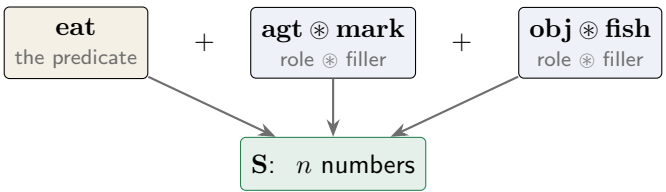
So agt ⊛ mark is unlike agt and unlike mark: a new item that means, roughly, *Mark as the agent*. And if mark and john were similar atoms, agt ⊛ mark and agt ⊛ john would be equally similar.

The last row is Session 11c's loss. Inside a binding, a role cannot be told from a filler. HRR therefore keeps roles in an inventory of their own, and argument order is carried by *which* role a filler is bound to.

Plate's Frames

A clause is a **frame**: its predicate, plus one binding per role, all added into one vector. Plate's own example (1995):

$$\textit{Mark ate the fish} \quad \mapsto \quad \mathbf{S} = \mathbf{eat} + \mathbf{agt} \circledast \mathbf{mark} + \mathbf{obj} \circledast \mathbf{fish}$$



Every term is a vector of length about 1, so $\mathbf{S}$ has length about $\sqrt{3}$, and it is *still* n numbers. A bigger frame adds more terms, never more dimensions.

Plate wrote predicate-specific roles, $\mathbf{agt}_{\text{eat}}$ and $\mathbf{obj}_{\text{eat}}$; we use one agt and one obj for all verbs, as in earlier sessions. Nothing below depends on the choice.

The question for Part II: given $\mathbf{S}$ and $\mathbf{agt}$, how do we get $\mathbf{mark}$ back?

Part II
Taking a Binding Apart

What Would Undo a Convolution?

Convolution has an identity: the vector $\boldsymbol{\delta} = (1, 0, \ldots, 0)$. In the formula only $k = 0$ survives:

$$(\boldsymbol{\delta} \circledast \mathbf{x})_j \;=\; \sum_k \delta_k \, x_{(j-k) \bmod n} \;=\; x_j, \qquad \text{so} \qquad \boldsymbol{\delta} \circledast \mathbf{x} = \mathbf{x}$$

An **inverse** of $\mathbf{c}$ would be a vector $\mathbf{c}^{-1}$ with $\mathbf{c}^{-1} \circledast \mathbf{c} = \boldsymbol{\delta}$. Then unbinding would be exact:

$$\mathbf{c}^{-1} \circledast (\mathbf{c} \circledast \mathbf{x}) \;=\; (\mathbf{c}^{-1} \circledast \mathbf{c}) \circledast \mathbf{x} \;=\; \boldsymbol{\delta} \circledast \mathbf{x} \;=\; \mathbf{x}$$

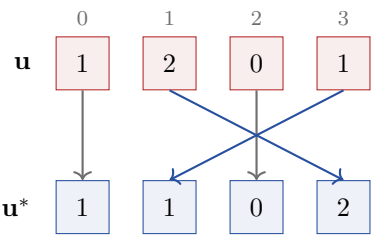
Most vectors have an inverse, but computing it means dividing, and Part V shows why dividing is dangerous once a vector holds several bindings.

Plate's choice is cheaper and more robust: an **approximate** inverse that needs no arithmetic at all, only a reordering of coordinates. It is called the **involution**.

The Involution, Step by Step

Keep the first coordinate and reverse the rest:

$$\mathbf{c}^* \;=\; (c_0,\; c_{n-1},\; c_{n-2},\; \ldots,\; c_1), \qquad \text{that is,} \qquad c_i^* = c_{(-i) \bmod n}$$



For $\mathbf{u} = (1, 2, 0, 1)$: position 0 stays, positions 1 and 3 trade places, and position 2, its own mirror image mod 4, stays:

$$\mathbf{u}^* = (1,\; 1,\; 0,\; 2)$$

Do it twice and $\mathbf{u}$ comes back, $(\mathbf{u}^*)^* = \mathbf{u}$: that is what makes it an *involution*.

You have met this vector before. In 11c's fold, the first step reversed $\mathbf{v}$ and rotated it once, giving $(2, 1, 1, 0)$ – which is $\mathbf{v}^*$.

Correlation: Sliding One Vector Along Another

Plate unbinds by convolving with the involution. The operation has its own name, **circular correlation**, written $\#$:

$$\mathbf{c} \# \mathbf{t} \;=\; \mathbf{c}^* \circledast \mathbf{t}, \qquad \text{written out:} \qquad (\mathbf{c} \# \mathbf{t})_j \;=\; \sum_k c_k \, t_{(k+j) \bmod n}$$

Read the right-hand formula as a recipe: rotate $\mathbf{t}$ left by j places, then take its dot product with $\mathbf{c}$. The $+j$ in the index is what rotates left: c_k meets t_{k+j}, the entry j places further along, so t_j comes to sit under c_0: for $j = 1$, $\mathbf{c}$ meets (t_1, t_2, t_3, t_0). Entry j measures how well $\mathbf{c}$ matches $\mathbf{t}$ at offset j; entry 0 is the plain dot product $\mathbf{c} \cdot \mathbf{t}$.

	convolution $\mathbf{c} \circledast \mathbf{t}$	**correlation $\mathbf{c} \# \mathbf{t}$**
indices in each product	add to j: $c_k \, t_{j-k}$	differ by j: $c_k \, t_{k+j}$
job in HRR	bind	unbind
order of arguments	does not matter	matters: $\mathbf{c} \# \mathbf{t} \neq \mathbf{t} \# \mathbf{c}$ in general

The name comes from signal processing, where correlating a known pattern against a signal finds where the pattern occurs.

Why $+j$ in the Index Rotates Left

Build a new vector $\mathbf{s}$ whose entry k is t_{k+j}: each slot reads the entry j *places ahead*. With $\mathbf{t} = (4, 5, 2, 5)$ and $j = 1$:

slot k	0	1	2	3
reads index $k + 1 \pmod 4$	1	2	3	0
$s_k = t_{k+1}$	5	2	5	4

So $\mathbf{s} = (5, 2, 5, 4)$: every entry of $\mathbf{t}$ has moved one place toward the front, and t_0 wrapped around to the back. Each t_m now sits in slot $m - j$. Reading ahead moves the contents left.

index	**each slot reads**	**the contents move**
t_{k+j} (correlation)	j places ahead	j places left: t_j lands in slot 0
t_{k-j} (convolution, the shift ρ^j)	j places behind	j places right: t_0 lands in slot j

A sentence makes it concrete. Store *the dog chased the cat* by position, and let each slot read the word one place ahead: *dog chased the cat the*. The sentence has moved one word left. Correlation then dots $\mathbf{c}$ with $\mathbf{s}$; for $\mathbf{c} = \mathbf{u}$, $\mathbf{u} \cdot (5, 2, 5, 4) = 5 + 4 + 0 + 4 = 13 = (\mathbf{u} \# \mathbf{t})_1$.

Why Convolving With the Involution Slides

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Write out $\mathbf{c}^* \circledast \mathbf{t}$ with the convolution formula, use $c_k^* = c_{-k}$, and rename $m = -k$ (all indices mod n):

$$(\mathbf{c}^* \circledast \mathbf{t})_j \;=\; \sum_k c_k^*\, t_{j-k} \;=\; \sum_k c_{-k}\, t_{j-k} \;=\; \sum_m c_m\, t_{j+m} \;=\; (\mathbf{c} \,\#\, \mathbf{t})_j$$

As k runs over every index, so does m. Check it with $\mathbf{u}^* = (1, 1, 0, 2)$ and $\mathbf{t} = \mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5)$:

j	**convolve $\mathbf{u}^*$ with $\mathbf{t}$** (indices add to j)	**slide $\mathbf{u}$ along $\mathbf{t}$** (indices differ by j)
0	$1{\cdot}4 + 1{\cdot}5 + 0{\cdot}2 + 2{\cdot}5 = 19$	$1{\cdot}4 + 2{\cdot}5 + 0{\cdot}2 + 1{\cdot}5 = 19$
1	$1{\cdot}5 + 1{\cdot}4 + 0{\cdot}5 + 2{\cdot}2 = 13$	$1{\cdot}5 + 2{\cdot}2 + 0{\cdot}5 + 1{\cdot}4 = 13$

The same four products appear in both columns, in a different order. The involution only re-indexes the role, so that convolution's "indices add to j" becomes correlation's "indices differ by j." Plate gives $\#$ its own symbol, but it is always convolution with the involution.

Decoding by Hand

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Bind, $\mathbf{t} = \mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5)$. To recover $\mathbf{v}$, correlate with $\mathbf{u}$: rotate $\mathbf{t}$ left by j and dot with $\mathbf{u}$.

j	$\mathbf{t}$ rotated left by j, dotted with $\mathbf{u} = (1, 2, 0, 1)$	$(\mathbf{u} \# \mathbf{t})_j$
0	$(4, 5, 2, 5) \cdot \mathbf{u} = 4 + 10 + 0 + 5$	19
1	$(5, 2, 5, 4) \cdot \mathbf{u} = 5 + 4 + 0 + 4$	13
2	$(2, 5, 4, 5) \cdot \mathbf{u} = 2 + 10 + 0 + 5$	17
3	$(5, 4, 5, 2) \cdot \mathbf{u} = 5 + 8 + 0 + 2$	15

So $\mathbf{u} \# (\mathbf{u} \circledast \mathbf{v}) = (19, 13, 17, 15)$. Compare it with the candidates:

$$\cos\big((19, 13, 17, 15),\ \mathbf{v}\big) = 0.88, \qquad \cos\big((19, 13, 17, 15),\ \mathbf{u}\big) = 0.76$$

Closer to $\mathbf{v}$ than to anything else here – but it is not $\mathbf{v}$. At $n = 4$ everything overlaps ($\mathbf{u}$ and $\mathbf{v}$ themselves sit at 0.50), so the margin is thin. The next slide finds the error exactly.

Where the Error Comes From

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Convolution is commutative and associative, so the decoding can be regrouped:

$$\mathbf{u} \# (\mathbf{u} \circledast \mathbf{v}) \;=\; \mathbf{u}^* \circledast \mathbf{u} \circledast \mathbf{v} \;=\; (\mathbf{u} \# \mathbf{u}) \circledast \mathbf{v}$$

$\mathbf{u} \# \mathbf{u}$ is $\mathbf{u}$ correlated with itself, its **autocorrelation**. Entry j dots $\mathbf{u}$ with itself shifted by j:

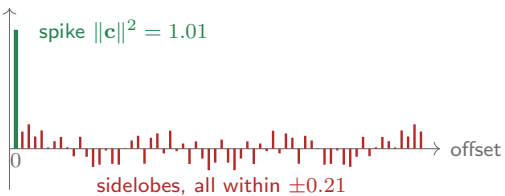
$$\mathbf{u} \# \mathbf{u} = (\underbrace{6}_{\mathbf{u}\cdot\mathbf{u}},\ \underbrace{3,\ 4,\ 3}_{\text{sidelobes}}) \;=\; 6\,\boldsymbol{\delta} \;+\; (0, 3, 4, 3)$$

The spike at offset 0 is $\|\mathbf{u}\|^2$; the sidelobes are the overlaps at every other offset. Convolve with $\mathbf{v}$:

$$\mathbf{u} \# (\mathbf{u} \circledast \mathbf{v}) \;=\; \underbrace{6\,\mathbf{v}}_{(12,0,6,6)} \;+\; \underbrace{(0, 3, 4, 3) \circledast \mathbf{v}}_{(7,13,11,9)} \;=\; (19, 13, 17, 15)\ \checkmark$$

Decoding returns $\|\mathbf{u}\|^2\,\mathbf{v}$ plus the sidelobes convolved with $\mathbf{v}$: exact only when every sidelobe is zero.

At Large n the Sidelobes Are Small, but Many



the autocorrelation $\mathbf{c} \# \mathbf{c}$ of one random atom, $n = 64$

For a random atom at $n = 1{,}024$, measured over 300 atoms:

- spike: 1.00 ± 0.04
- each sidelobe: about ± 0.03, that is $\pm 1/\sqrt{n}$
- but there are $1{,}023$ of them, with total energy 1.00

So the noise in $\mathbf{c} \# (\mathbf{c} \circledast \mathbf{x}) = \mathbf{x} + \text{noise}$ is about as long as $\mathbf{x}$ itself, and points in a random direction. Two equal lengths at right angles give

$$\cos\big(\mathbf{c} \# (\mathbf{c} \circledast \mathbf{x}),\ \mathbf{x}\big) \ \approx\ \frac{1}{\sqrt{1+1}} \ =\ 0.71 \qquad \text{measured: } 0.71 \pm 0.02$$

Against any other atom the decoded vector scores 0.00 ± 0.03. Recognizable, not exact: 0.71 against a floor of 0.03.

Unitary Vectors: Decoding Without Error

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Some vectors have no sidelobes at all. Take $\mathbf{r} = \frac{1}{2}(1, 1, 1, -1)$: its autocorrelation is $\mathbf{r} \# \mathbf{r} = (1, 0, 0, 0) = \boldsymbol{\delta}$. Bind $\mathbf{v}$ to it, then unbind with $\mathbf{r}^* = \frac{1}{2}(1, -1, 1, 1)$:

bind: $\mathbf{r} \circledast \mathbf{v}$		
0	$\frac{1}{2}(2+1+1-0)$	2
1	$\frac{1}{2}(0+2+1-1)$	1
2	$\frac{1}{2}(1+0+2-1)$	1
3	$\frac{1}{2}(1+1+0-2)$	0

unbind: $\mathbf{r}^* \circledast (2,1,1,0)$		
0	$\frac{1}{2}(2-0+1+1)$	2
1	$\frac{1}{2}(1-2+0+1)$	0
2	$\frac{1}{2}(1-1+2+0)$	1
3	$\frac{1}{2}(0-1+1+2)$	1

$\mathbf{r} \# (\mathbf{r} \circledast \mathbf{v}) = (2, 0, 1, 1) = \mathbf{v}$, exactly. Each row is the convolution formula, $(\mathbf{a} \circledast \mathbf{b})_j = \sum_k a_k \, b_{(j-k) \bmod n}$, with $\mathbf{a} = \mathbf{r}$ in the first table and $\mathbf{a} = \mathbf{r}^*$ in the second.

Such vectors are called **unitary**. For them the involution *is* the exact inverse, $\mathbf{r}^* = \mathbf{r}^{-1}$. Part V gives the test: all of $\mathbf{r}$'s Fourier coefficients have size 1.

The Cyclic Shift Is a Unitary Binding

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

The simplest unitary vector is a unit impulse one place along, $\boldsymbol{\delta}_1 = (0, 1, 0, 0)$. Convolving with it moves every coordinate one place right, and the last one to the front:

$$\boldsymbol{\delta}_1 \circledast \mathbf{v} \;=\; (v_3,\; v_0,\; v_1,\; v_2) \;=\; (1,\; 2,\; 0,\; 1)$$

Its involution is $\boldsymbol{\delta}_1^* = (0, 0, 0, 1)$, which shifts one place left, so the shift is undone exactly.

The cyclic shift ρ that the next session uses to mark word order is convolution with a unit impulse: a special HRR binding, unitary, and so exactly invertible. Shifting i places is binding with $\boldsymbol{\delta}_i$.

A small surprise in our running example: $\boldsymbol{\delta}_1 \circledast \mathbf{v} = (1, 2, 0, 1) = \mathbf{u}$. Our $\mathbf{u}$ is $\mathbf{v}$ shifted by one place. Exercise 1 uses this.

Why Plate Used the Involution Anyway

An exact inverse exists for most atoms. Compare it with the involution on traces that hold k role–filler pairs, decoding one of them, at $n = 1{,}024$:

pairs in the trace, k	1	2	3	5	10
exact inverse $\mathbf{c}^{-1}$	1.00	0.34	0.25	0.18	0.12
involution $\mathbf{c}^{*}$	0.71	0.58	0.50	0.41	0.30

Mean cosine to the correct filler over 300 traces.

With one pair the exact inverse is perfect. Add a second pair and it falls below the involution, and stays there. The reason, in Part V: the exact inverse *divides* by the atom's Fourier coefficients, and a random atom always has a few small ones (the smallest is about 0.04 at this n). Whatever else sits in the trace at those frequencies is multiplied by up to about 28.

Plate notes that inverse filters are sensitive to noise, so on superposed traces the involution gives the better signal-to-noise ratio – and for unitary atoms the two coincide.

In-Class Exercise 1

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

1. Compute $\mathbf{v}^*$, then decode the other way: $\mathbf{v} \# (\mathbf{u} \circledast \mathbf{v})$. Which of $\mathbf{u}$, $\mathbf{v}$ is the result closest to?
2. Compute $\mathbf{v} \# \mathbf{v}$, the autocorrelation of $\mathbf{v}$. Compare it with $\mathbf{u} \# \mathbf{u} = (6, 3, 4, 3)$, and explain the match.
3. Show that $\frac{1}{2}(1, -1, 1, 1)$ is unitary by computing its autocorrelation. How is it related to $\mathbf{r} = \frac{1}{2}(1, 1, 1, -1)$?
4. $\star$ Decoding gave $6\,\mathbf{v}$ plus noise rather than $\mathbf{v}$ plus noise. Where does the 6 come from, and what does Plate's variance $1/n$ buy?

Seven minutes. Item 4 is the starred one.

In-Class Exercise 1: Answers

1. $\mathbf{v}^* = (2, 1, 1, 0)$. The bound pair is $\mathbf{t} = \mathbf{u} \circledast \mathbf{v} = (4, 5, 2, 5)$, as in Decoding by Hand. Rotating $\mathbf{t}$ left by j and dotting with $\mathbf{v}$ gives $\mathbf{v} \# \mathbf{t} = (15, 19, 13, 17)$, at cosine 0.88 with $\mathbf{u}$. Either factor of a binding decodes the other, because the binding is commutative.

2. $\mathbf{v} \# \mathbf{v} = (6, 3, 4, 3)$, the same as $\mathbf{u}$'s. Since $\mathbf{u} = \boldsymbol{\delta}_1 \circledast \mathbf{v}$ is $\mathbf{v}$ shifted, every offset compares the same pairs of numbers. A shift never changes an autocorrelation.

3. Its autocorrelation is $\frac{1}{4}(4, 0, 0, 0) = \boldsymbol{\delta}$: offset 0 gives $\frac{1}{4}(1 + 1 + 1 + 1)$, offset 1 gives $\frac{1}{4}(-1 - 1 + 1 + 1)$, offset 2 gives $\frac{1}{4}(1 - 1 + 1 - 1)$, and offset 3 matches offset 1, since an autocorrelation is symmetric. It is $\mathbf{r}^*$: the involution of a unitary vector is unitary.

4. $\star$ The spike is $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2 = 6$. With variance $1/n$ every atom has $\|\mathbf{c}\|^2 \approx 1$, so decoding returns the filler at its own scale, cleanup can compare it directly with stored atoms, and the terms of a frame stay equally weighted.

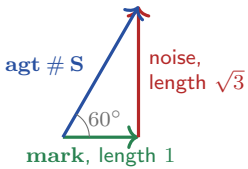
Part III
Frames at Work

Querying a Frame, Term by Term

Ask *Mark ate the fish* who did the eating: correlate the frame with $\mathbf{agt}$. Correlation distributes over the sum, term by term:

$$\mathbf{agt} \,\#\, \mathbf{S} \;=\; \underbrace{\mathbf{agt} \,\#\, \mathbf{eat}}_{\text{noise}} \;+\; \underbrace{\mathbf{agt} \,\#\, (\mathbf{agt} \circledast \mathbf{mark})}_{\mathbf{mark} \,+\, \text{sidelobe noise}} \;+\; \underbrace{\mathbf{agt} \,\#\, (\mathbf{obj} \circledast \mathbf{fish})}_{\text{noise}}$$

Count the pieces. One copy of $\mathbf{mark}$, of length about 1.
Three noise pieces, each of length about 1 and pointing in unrelated directions, so together of length about $\sqrt{3}$, at right angles to $\mathbf{mark}$:



$$\cos(\mathbf{agt} \,\#\, \mathbf{S},\ \mathbf{mark}) \;\approx\; \frac{1}{\sqrt{1+3}} \;=\; 0.50$$

Measured: 0.50, over 400 frames at $n = 1{,}024$.
The noise is only nearly perpendicular to $\mathbf{mark}$, as any random direction is at large n. Total length 2, so $\cos 60° = 1/2$.

Cleanup Returns the Filler

The decoded vector is not an atom. A **cleanup memory** holds every atom we minted and returns the one with the highest cosine. One frame, $n = 1{,}024$, nine atoms in the memory:

question	compute	best atom	runner-up
who ate?	$\mathbf{agt} \# \mathbf{S}$	mark 0.47	0.04
what was eaten?	$\mathbf{obj} \# \mathbf{S}$	fish 0.49	0.03
what role does the fish fill?	$\mathbf{fish} \# \mathbf{S}$	obj 0.49	0.06
what role does Mark fill?	$\mathbf{mark} \# \mathbf{S}$	agt 0.46	0.06

Atoms in the memory: eat, see, cause, mark, john, fish, hunger, agt, obj. Over 400 frames the right atom averages 0.50; every other atom, 0.00 ± 0.03.

Unbinding works in both directions: by a role, to get its filler, or by a filler, to get its role, because the binding is commutative.

Cleanup is where the symbol comes back. Everything before it is continuous; cleanup rounds to a finite list. The next session returns to what that rounding means.

Order Lives in the Roles

Compare *Mark ate the fish*, $\mathbf{S}_1 = \mathbf{eat} + \mathbf{agt} \circledast \mathbf{mark} + \mathbf{obj} \circledast \mathbf{fish}$, with other frames built the same way:

frame	terms shared with $\mathbf{S}_1$	$\cos(\mathbf{S}_1, \cdot)$	bag of words
The fish ate Mark	$\mathbf{eat}$	0.33	1.00
John ate the fish	$\mathbf{eat}$, $\mathbf{obj} \circledast \mathbf{fish}$	0.67	0.67
Mark saw the fish	$\mathbf{agt} \circledast \mathbf{mark}$, $\mathbf{obj} \circledast \mathbf{fish}$	0.67	0.67
John saw Mark	none	0.00	0.33

Mean over 400 draws at $n = 1{,}024$. Each frame has three terms of length about 1, so sharing s of them gives a cosine of about $s/3$.

HRR similarity counts shared role–filler pairs, not shared words. *The fish ate Mark* has every word of *Mark ate the fish*, so a bag of words scores them as identical; HRR keeps only the predicate in common, because mark and fish have traded roles.

This is Session 11c's diagnosis, answered the HRR way: the binding cannot see order, but the choice of role can.

Part IV
How Much Fits

Counting the Noise

Decode one pair from a trace that holds k pairs. The signal is one filler; the noise has k pieces, the other $k-1$ pairs plus the decoding role's own sidelobes. All of them have length about 1, and unrelated vectors add like the sides of a right triangle, so $\cos \approx 1/\sqrt{1 + \text{number of noise pieces}}$:

$$\cos \approx \frac{1}{\sqrt{k+1}} \qquad \text{and, with unitary roles (no sidelobes),} \qquad \cos \approx \frac{1}{\sqrt{k}}$$

pairs in the trace, k	1	2	3	5	9	20
random roles, measured	0.71	0.58	0.50	0.41	0.31	0.22
$1/\sqrt{k+1}$	0.71	0.58	0.50	0.41	0.32	0.22
unitary roles, measured	1.00	0.71	0.58	0.45	0.33	0.22
$1/\sqrt{k}$	1.00	0.71	0.58	0.45	0.33	0.22

$n = 1{,}024$, mean of 200 traces. A frame's predicate counts as one more term, which is why *Mark ate the fish* gave 0.50: three terms, so $k = 3$.

Unitary roles help most when the trace is small; by $k = 20$ the other pairs swamp them. Session 11c's *John loves Mary* trace had no predicate, only $\mathbf{agt} \circledast \mathbf{john} + \mathbf{pat} \circledast \mathbf{mary}$: $\mathbf{john}$ is the signal, and $\mathbf{agt}$'s sidelobes and $\mathbf{pat} \circledast \mathbf{mary}$ are two noise pieces, so $1/\sqrt{1+2} = 0.58$ (11c measured 0.599).

When Cleanup Fails

Cleanup fails when some wrong atom outscores the right one. Each wrong atom scores about $\pm 1/\sqrt{n}$, and the best of m wrong atoms reaches about

$$\frac{\sqrt{2\ln m}}{\sqrt{n}} \qquad \text{for } m = 1{,}000 \text{ and } n = 1{,}024\text{:} \quad \frac{\sqrt{13.8}}{32} \approx 0.12$$

The right atom scores about $1/\sqrt{k+1}$. Trouble starts when the two meet:

$$\frac{1}{\sqrt{k+1}} \approx \frac{\sqrt{2\ln m}}{\sqrt{n}} \iff k+1 \approx \frac{n}{2\ln m} \implies k \approx 70 \text{ at } n = 1{,}024$$

The dimension needed grows linearly with the number of pairs, and only logarithmically with the size of the cleanup memory. Plate (1995) derives the same shape with care; this back-of-the-envelope version gets the scaling right.

The $\sqrt{2\ln m}$ is the usual size of the largest of m independent standard normal scores. Doubling the vocabulary barely moves it; doubling the number of pairs needs twice the dimension.

Measured Capacity

Bind k random role–filler pairs into one trace, decode the first role, and clean up against $m = 1{,}000$ fillers. Fraction of trials returning the right filler:

n	$k = 1$	2	5	10	20	50	100
256	1.00	1.00	1.00	0.93	0.61	0.18	0.05
1,024	1.00	1.00	1.00	1.00	1.00	0.85	0.46
4,096	1.00	1.00	1.00	1.00	1.00	1.00	1.00

150 trials per cell.

At $n = 1{,}024$ the first failures appear between $k = 20$ and $k = 50$, and half the trials fail near $k = 100$; the estimate of about 70 from the last slide falls in between. Quadrupling n moves the breaking point about four times further out: at $n = 256$, half the trials fail near $k = 25$.

The failure is a cliff, not a slope. Size n for the largest structure you intend to store.

Nesting: *Hunger Caused Mark to Eat the Fish*

Plate's recursive example. A whole frame can fill a role, because it is just another vector of n numbers:

$$\mathbf{E} = \mathbf{eat} + \mathbf{agt} \circledast \mathbf{mark} + \mathbf{obj} \circledast \mathbf{fish}, \qquad \mathbf{C} = \mathbf{cause} + \mathbf{agt} \circledast \mathbf{hunger} + \mathbf{obj} \circledast \mathbf{E}$$

Who ate? needs two decodings: first $\mathbf{obj} \# \mathbf{C}$, to reach the caused event, then $\mathbf{agt}$ on the result. Measured at $n = 1{,}024$:

step	compared with	cos
$\mathbf{obj} \# \mathbf{C}$	$\mathbf{E}$	0.67
$\mathbf{agt} \# (\mathbf{obj} \# \mathbf{C})$, no cleanup in between	$\mathbf{mark}$	0.26
clean up $\mathbf{obj} \# \mathbf{C}$ to the stored $\mathbf{E}$, then decode with $\mathbf{agt}$	$\mathbf{mark}$	0.50

Cleanup chose $\mathbf{E}$ over three other stored frames in all 400 trials.

Clean up at every level. The noise of two decodings compounds; cleanup in between resets it. That requires $\mathbf{E}$ itself to sit in cleanup memory, as one item. Hinton (1990) called such a stored, compressed whole a **reduced description**, and it is the "reduced" in Plate's name.

Sequences, and the Shift Again

Order within a string can be carried by powers of one unitary key $\mathbf{p}$, written $\mathbf{p}^2 = \mathbf{p} \circledast \mathbf{p}$ and so on:

$$\textit{the dog barked} \quad \mapsto \quad \mathbf{p} \circledast \mathbf{the} \;+\; \mathbf{p}^2 \circledast \mathbf{dog} \;+\; \mathbf{p}^3 \circledast \mathbf{barked}$$

To read position 2, correlate with $\mathbf{p}^2$. Powers of a unitary vector are unitary, and different powers are nearly orthogonal, so each position decodes as one term among three. Measured at $n = 1{,}024$:

query	cos to the	cos to dog	cos to barked
$\mathbf{p} \# \mathbf{s}$	0.58	0.00	0.00
$\mathbf{p}^2 \# \mathbf{s}$	0.00	0.58	0.00
$\mathbf{p}^3 \# \mathbf{s}$	0.00	0.00	0.58

$0.58 = 1/\sqrt{3}$. Off-diagonal entries are 0.00 ± 0.03.

Choose $\mathbf{p} = \boldsymbol{\delta}_1 = (0, 1, 0, \ldots, 0)$ and the powers are shifts: $\boldsymbol{\delta}_1^{\,i} \circledast \mathbf{x}$ moves $\mathbf{x}$ by i places. That is how the next session encodes word order, writing $\rho^i(\mathbf{x})$ for a shift by i places.

In-Class Exercise 2

1. A frame has a predicate and eight role bindings. Estimate the cosine between the correct filler and the vector you get by decoding one role.
2. Is cleanup safe for that frame at $n = 1{,}024$ with $1{,}000$ atoms in memory? At $n = 256$?
3. With $\mathbf{C} = \mathbf{cause} + \mathbf{agt} \circledast \mathbf{hunger} + \mathbf{obj} \circledast \mathbf{E}$ for *Hunger caused Mark to eat the fish*, write the queries that answer *What caused it?* and *What was eaten?* Say where cleanup is needed.
4. $\star$ Nested queries worked only because $\mathbf{E}$ was stored in cleanup memory. What does that say about where the symbols live in HRR?

Seven minutes. Item 4 is the starred one.

In-Class Exercise 2: Answers

1. The frame is a sum of **nine terms**: the predicate, plus eight role–filler bindings. Decode one role, $\mathbf{r} \# \text{frame}$, and every term contributes:

- from $\mathbf{r} \circledast \mathbf{f}$: the filler $\mathbf{f}$ itself, length about 1, plus $\mathbf{r}$'s own sidelobes, noise piece 1;
- from the predicate and the seven other bindings: noise pieces 2 to 9, each of length about 1.

The nine noise pieces point in unrelated directions, so their lengths add as squares: total noise $\sqrt{9}$. The decoded vector is the filler plus that noise, so its length is $\sqrt{1+9}$, and its cosine with the filler is the filler's share: $\cos \approx 1/\sqrt{9+1} \approx 0.32$. The $+1$ is the filler, not one more term: k terms give k noise pieces, and the signal adds 1 under the root. Measured: 0.31.

2. The best wrong atom reaches about $\sqrt{2 \ln 1000}/\sqrt{n}$: 0.12 at $n = 1{,}024$, comfortably below 0.32. At $n = 256$ it is about 0.23, too close for comfort; the capacity table shows about 7% failures at $k = 10$.

3. *What caused it?*: clean up $\mathbf{agt} \# \mathbf{C}$, which returns hunger. *What was eaten?*: clean up $\mathbf{obj} \# \mathbf{C}$ against stored frames, which returns $\mathbf{E}$; then clean up $\mathbf{obj} \# \mathbf{E}$, which returns fish. Two levels, two cleanups.

4. $\star$ In the cleanup memory: the list of atoms and of stored frames, which the next session calls the *codebook*. The vectors and the operations are continuous; the system becomes symbolic each time it rounds to that list. The next session ends on this claim.

Part V

Fourier, From Scratch

why convolution has a fast version, and what it says about decoding

The Question: Is There a Faster Way to Bind?

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Computed from the formula, each of the n outputs of $\mathbf{u} \circledast \mathbf{v}$ adds up n products, so one binding costs n^2 multiplications: 16 at $n = 4$, about a million at $n = 1{,}024$, about 17 million at $n = 4{,}096$. Part IV's capacity table needed n in the thousands, and every frame takes one binding per role.

Session 11c stated a faster route in one line, $\mathbf{u} \circledast \mathbf{v} = \mathcal{F}^{-1}(\mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v})$. This part builds that line from nothing, in four steps:

1. Convolution is a sum of shifted copies of one vector.
2. Some vectors come out of a shift as a multiple of themselves: **waves**.
3. So convolution multiplies each wave by a single number, a **Fourier coefficient**.
4. The **fast Fourier transform** (FFT) finds all n coefficients in about $(n/2)\log_2 n$ steps.

The same coefficients then settle two questions Part II left open: which roles decode exactly, and why the exact inverse does badly on crowded traces.

Step 1: Convolution Is a Sum of Shifted Copies

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Write ρ for the cyclic shift, which moves every coordinate one place right and the last to the front: $\rho(x_0, x_1, x_2, x_3) = (x_3, x_0, x_1, x_2)$. Shifting k places is ρ^k. Collect the terms of $(\mathbf{u} \circledast \mathbf{v})_j = \sum_k u_k \, v_{(j-k) \bmod n}$ by u_k instead of by j, and each u_k multiplies $\mathbf{v}$ shifted k places:

$$\mathbf{u} \circledast \mathbf{v} \;=\; u_0\,\mathbf{v} \;+\; u_1\,\rho(\mathbf{v}) \;+\; u_2\,\rho^2(\mathbf{v}) \;+\; u_3\,\rho^3(\mathbf{v})$$

weight	**shifted copy of $\mathbf{v}$**	**weight $\times$ copy**
$u_0 = 1$	$\mathbf{v} = (2, 0, 1, 1)$	$(2, 0, 1, 1)$
$u_1 = 2$	$\rho(\mathbf{v}) = (1, 2, 0, 1)$	$(2, 4, 0, 2)$
$u_2 = 0$	$\rho^2(\mathbf{v}) = (1, 1, 2, 0)$	$(0, 0, 0, 0)$
$u_3 = 1$	$\rho^3(\mathbf{v}) = (0, 1, 1, 2)$	$(0, 1, 1, 2)$
	sum	$(4, 5, 2, 5)$

The same $(4, 5, 2, 5)$ as the formula. Convolution does to $\mathbf{v}$ whatever shifting does, weighted and summed: if a shift only multiplies some vector by a number, so does convolution.

Step 2: Vectors That a Shift Only Rescales

Two are easy to find:

$$\rho(1,\ 1,\ 1,\ 1) = (1,\ 1,\ 1,\ 1) = 1 \cdot (1,\ 1,\ 1,\ 1)$$
$$\rho(1,\ -1,\ 1,\ -1) = (-1,\ 1,\ -1,\ 1) = -1 \cdot (1,\ -1,\ 1,\ -1)$$

Both have the form $(1,\ z,\ z^2,\ z^3)$: each entry is z times the one before. Shifting such a pattern brings z^3 to the front:

$$\rho(1,\ z,\ z^2,\ z^3) \ = \ (z^3,\ 1,\ z,\ z^2) \ = \ z^3 \cdot (1,\ z,\ z^2,\ z^3) \qquad \text{provided } z^4 = 1$$

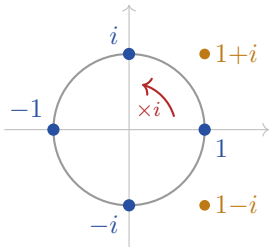
The condition is the wrap-around: the new second entry, 1, has to equal $z^3 \cdot z$.

Among real numbers, only $z = 1$ and $z = -1$ satisfy $z^4 = 1$. A basis for vectors of length 4 needs four such patterns, so two are missing. They need a number with $z^2 = -1$, which makes $z^4 = 1$ as well. No real number squares to -1, so we use the number that does: i.

The pattern needs a number that turns: after four steps it must be back at 1. The next slide collects the facts about i that this part uses.

A Minute on Complex Numbers

- A **complex number** $a + bi$ is a point in the plane: a across, b up.
- $i^2 = -1$, so the powers of i repeat every four steps: $1,\ i,\ -1,\ -i,\ 1,\ \ldots$ Multiplying by i is a quarter turn counterclockwise; multiplying by $-i$, a quarter turn back.
- Multiply as with polynomials, then replace i^2 by -1: $(1-i)(1+i) = 1 + i - i - i^2 = 2$.
- The **conjugate** $\overline{a+bi} = a - bi$ is the mirror image across the horizontal axis, and $z\,\overline{z} = a^2 + b^2 = |z|^2$, the squared distance from 0: $|1-i|^2 = 2$.



For general n the quarter turn becomes a $1/n$ turn: the point $e^{2\pi i/n} = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ on the circle of radius 1. At $n = 4$ that point is i. That is all the complex arithmetic this part uses.

The Four Waves at $n = 4$

Take $z = 1, i, -1, -i$ in turn. **Wave** k is $\mathbf{f}_k = (1,\ i^k,\ i^{2k},\ i^{3k})$: start at 1 and turn k quarter turns at every step. Each arrow shows one entry as a point on the circle:

wave	step 0	1	2	3	entries	turn per step	a shift multiplies it by
$\mathbf{f}_0$					$(1,\ 1,\ 1,\ 1)$	none	1
$\mathbf{f}_1$					$(1,\ i,\ -1,\ -i)$	a quarter	$-i$
$\mathbf{f}_2$					$(1,\ -1,\ 1,\ -1)$	a half	-1
$\mathbf{f}_3$					$(1,\ -i,\ -1,\ i)$	a quarter back	i

Check one: $\rho(1,\ i,\ -1,\ -i) = (-i,\ 1,\ i,\ -1) = -i \cdot (1,\ i,\ -1,\ -i)$. A shift never scrambles a wave: every arrow turns by the same amount, so the whole wave is multiplied by one number, $(-i)^k$ for wave k.

Wave 3 is wave 1 run backward: $\mathbf{f}_3 = \overline{\mathbf{f}_1}$. The four waves form a basis, as $\mathbf{e}_0, \ldots, \mathbf{e}_3$ do, but made of patterns instead of positions.

Step 3: Convolution Multiplies Each Wave by One Number

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Put Steps 1 and 2 together. Convolving with $\mathbf{u}$ adds up shifted copies, and every shifted copy of a wave is the wave times a number:

$$\begin{aligned}\mathbf{u} \circledast \mathbf{f}_k &= u_0\,\mathbf{f}_k + u_1\,\rho(\mathbf{f}_k) + u_2\,\rho^2(\mathbf{f}_k) + u_3\,\rho^3(\mathbf{f}_k) \\ &= \left[\, u_0 + u_1(-i)^k + u_2(-i)^{2k} + u_3(-i)^{3k} \,\right] \mathbf{f}_k\end{aligned}$$

The bracket is one number for each k: the kth **Fourier coefficient** of $\mathbf{u}$,

$$(\mathcal{F}\mathbf{u})_k \;=\; \sum_j u_j\,(-i)^{jk}$$

Check at $k=1$: $(\mathcal{F}\mathbf{u})_1 = 1 - 2i - 0 + i = 1 - i$. Convolving directly, $\mathbf{u} \circledast \mathbf{f}_1 = (1{-}i,\ 1{+}i,\ -1{+}i,\ -1{-}i)$, which is $(1-i)\,\mathbf{f}_1$: the same wave, times $1-i$.

Fourier coefficient k of $\mathbf{u}$ is the number that convolving with $\mathbf{u}$ multiplies wave k by. For general n the recipe is the same with a $1/n$ turn: $(\mathcal{F}\mathbf{x})_k = \sum_j x_j\, e^{-2\pi i jk/n}$.

Computing the Coefficients, Row by Row

Coefficient k is the dot product of $\mathbf{x}$ with $\big(1,\ (-i)^k,\ (-i)^{2k},\ (-i)^{3k}\big)$, which is wave k turned backward. Stacked, these rows form the matrix $\mathcal{F}$:

$$\mathcal{F} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}$$

k	$(\mathcal{F}\mathbf{u})_k,\ \ \mathbf{u} = (1,2,0,1)$	$(\mathcal{F}\mathbf{v})_k,\ \ \mathbf{v} = (2,0,1,1)$
0	$1+2+0+1=4$	$2+0+1+1=4$
1	$1-2i-0+i=1-i$	$2-0-1+i=1+i$
2	$1-2+0-1=-2$	$2-0+1-1=2$
3	$1+2i-0-i=1+i$	$2+0-1-i=1-i$

So $\mathcal{F}\mathbf{u} = (4,\ 1{-}i,\ -2,\ 1{+}i)$ and $\mathcal{F}\mathbf{v} = (4,\ 1{+}i,\ 2,\ 1{-}i)$; the list $\mathcal{F}\mathbf{x}$ is the **spectrum** of $\mathbf{x}$. Coefficient 0 is the plain sum, coefficient 2 the alternating sum, and for a real vector coefficient 3 is the conjugate of coefficient 1.

Row by row, the n coefficients take n^2 multiplications: no faster than convolving directly. Step 4 fixes that.

Nothing Is Lost: Putting the Waves Back Together

The coefficients say how much of each wave a vector contains: weight wave k by coefficient k, add, and divide by $n = 4$. For $\mathbf{v}$, with $\mathcal{F}\mathbf{v} = (4,\ 1{+}i,\ 2,\ 1{-}i)$:

	index 0	**index 1**	**index 2**	**index 3**
$4\,\mathbf{f}_0$	4	4	4	4
$(1{+}i)\,\mathbf{f}_1$	$1{+}i$	$-1{+}i$	$-1{-}i$	$1{-}i$
$2\,\mathbf{f}_2$	2	-2	2	-2
$(1{-}i)\,\mathbf{f}_3$	$1{-}i$	$-1{-}i$	$-1{+}i$	$1{+}i$
sum	8	0	4	4
divided by 4	2	0	1	1

Back to $\mathbf{v} = (2, 0, 1, 1)$. The imaginary parts cancel in pairs, because terms 1 and 3 are conjugates.

$\mathcal{F}$ only re-describes a vector: n numbers in, n out. The **inverse transform** puts them back exactly, $x_j = \frac{1}{n}\sum_k (\mathcal{F}\mathbf{x})_k\, e^{+2\pi i j k/n}$: the same recipe with the turn reversed. The $\frac{1}{n}$ is there because the waves are perpendicular, each of squared length n (in the complex dot product, which conjugates one side).

The Convolution Theorem

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Now combine the steps. Write $\mathbf{v}$ as its mix of waves and convolve with $\mathbf{u}$. Convolution is linear, and it multiplies wave k by $(\mathcal{F}\mathbf{u})_k$ (Step 3):

$$\mathbf{u} \circledast \mathbf{v} \;=\; \tfrac{1}{4}\sum_k (\mathcal{F}\mathbf{v})_k \left(\mathbf{u} \circledast \mathbf{f}_k\right) \;=\; \tfrac{1}{4}\sum_k (\mathcal{F}\mathbf{u})_k \,(\mathcal{F}\mathbf{v})_k \,\mathbf{f}_k$$

So the coefficients of $\mathbf{u} \circledast \mathbf{v}$ are the products of the coefficients of $\mathbf{u}$ and $\mathbf{v}$:

$$\mathcal{F}(\mathbf{u} \circledast \mathbf{v}) \;=\; \mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v} \qquad \text{or, transforming back,} \qquad \mathbf{u} \circledast \mathbf{v} \;=\; \mathcal{F}^{-1}\big(\mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v}\big)$$

On our vectors: $(4,\ 1{-}i,\ -2,\ 1{+}i) \odot (4,\ 1{+}i,\ 2,\ 1{-}i) = (16,\ 2,\ -4,\ 2)$, using $(1-i)(1+i) = 2$. Back in ordinary coordinates, $\frac{1}{4}(16\,\mathbf{f}_0 + 2\,\mathbf{f}_1 - 4\,\mathbf{f}_2 + 2\,\mathbf{f}_3) = \frac{1}{4}(16, 20, 8, 20) = (4, 5, 2, 5)$, the binding from the formula.

The commutativity, in one glance: the middle step multiplies numbers, and numbers commute, so $\mathbf{u} \circledast \mathbf{v} = \mathbf{v} \circledast \mathbf{u}$. Session 11c's fold shows where the symmetry comes from: the antisymmetric half of $\mathbf{u}\mathbf{v}^\top$ is deleted.

Step 4: The FFT Splits the Work in Half

Row by row costs n^2. The **fast Fourier transform** (FFT) gets the same coefficients by splitting $\mathbf{x}$ into its even and odd positions. Since $(-i)^{2k} = (-1)^k$,

$$(\mathcal{F}\mathbf{x})_k \;=\; \underbrace{x_0 + (-1)^k\, x_2}_{E_k\text{: the evens}} \;+\; (-i)^k\; \underbrace{\left(x_1 + (-1)^k\, x_3\right)}_{O_k\text{: the odds}}$$

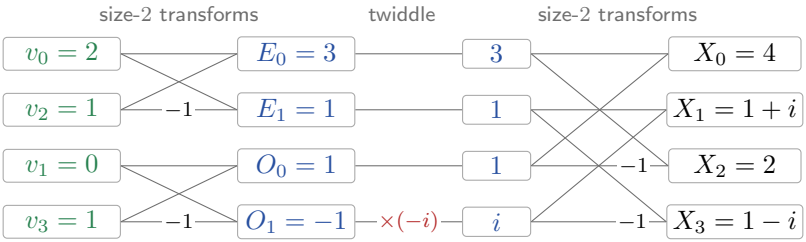
E and O are transforms of size 2, $(a, b) \mapsto (a + b,\ a - b)$. Since $(-i)^{k+2} = -(-i)^k$, the outputs come in pairs, $X_k = E_k + (-i)^k O_k$ and $X_{k+2} = E_k - (-i)^k O_k$ for $k = 0, 1$: each twiddled product serves two outputs.

step	**on $\mathbf{u} = (1, 2, 0, 1)$**
evens $(u_0, u_2) = (1, 0)$	$E = (1 + 0,\ 1 - 0) = (1,\ 1)$
odds $(u_1, u_3) = (2, 1)$	$O = (2 + 1,\ 2 - 1) = (3,\ 1)$
twiddle: multiply O_k by $(-i)^k$	3 and $-i$
combine	$X_0, X_2 = 1 \pm 3 = 4,\ -2$; $\quad X_1, X_3 = 1 \pm (-i) = 1{-}i,\ 1{+}i$

The same $\mathcal{F}\mathbf{u} = (4,\ 1{-}i,\ -2,\ 1{+}i)$ as row by row.

The FFT as a Diagram

The same steps for $\mathbf{v} = (2, 0, 1, 1)$, drawn as wires. Each pair of crossing wires is a size-2 transform, $(a, b) \mapsto (a + b,\ a - b)$; the label -1 marks the wire that is subtracted.



Evens $(2, 1) \mapsto (3, 1)$; odds $(0, 1) \mapsto (1, -1)$; the twiddle multiplies $O_1 = -1$ by $-i$, giving i; then $(3, 1) \mapsto (4, 2)$ and $(1, i) \mapsto (1+i,\ 1-i)$. That is $\mathcal{F}\mathbf{v}$, as row by row.

For $n = 8$, each half is itself a size-4 transform, split the same way: three stages of crossings instead of two. Every doubling of n adds one stage.

Why Halving Pays

Each split turns a transform of size n into two of size $n/2$, plus $n/2$ twiddle multiplications to combine them. Splitting again and again gives $\log_2 n$ stages of $n/2$ multiplications: about $(n/2)\log_2 n$ in all.

n	**binding, direct:** n^2	**one FFT:** $\frac{n}{2}\log_2 n$	**binding by FFT:** 3 FFTs + n
4	16	4	16
1,024	1,048,576	5,120	16,384
4,096	16,777,216	24,576	77,824
16,384	268,435,456	114,688	360,448

At $n = 4{,}096$, the largest size in Part IV's capacity table, a binding by FFT takes about $78{,}000$ multiplications instead of about 17 million, over 200 times fewer; at $n = 16{,}384$, about 750 times fewer. The counts are complex multiplications, without the additions; what matters is the growth, $n \log n$ against n^2. Fast algorithms exist for every n.

Where the algorithm comes from. Cooley and Tukey (1965) made it standard. Gauss had used the same splitting around 1805, computing the orbit of the asteroid Pallas, in notes published only after his death.

What the Coefficients Say About Decoding

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

The involution keeps the first entry and reverses the rest: $\mathbf{u}^* = (1, 1, 0, 2)$. Reversing the order runs every wave backward, so it **conjugates** every coefficient:

$$\mathcal{F}\mathbf{u}^* \;=\; (4,\; 1{+}i,\; -2,\; 1{-}i) \;=\; \overline{\mathcal{F}\mathbf{u}}$$

Decoding convolves with $\mathbf{u}^*$, so by the convolution theorem it multiplies coefficient k by $\overline{(\mathcal{F}\mathbf{u})_k}\,(\mathcal{F}\mathbf{u})_k = |(\mathcal{F}\mathbf{u})_k|^2$:

$$\mathcal{F}\big(\mathbf{u} \,\#\, (\mathbf{u} \circledast \mathbf{v})\big) \;=\; |\mathcal{F}\mathbf{u}|^2 \odot \mathcal{F}\mathbf{v}, \qquad |\mathcal{F}\mathbf{u}|^2 = (16,\; 2,\; 4,\; 2)$$

and $(16, 2, 4, 2)$ transforms back to $(6, 3, 4, 3)$, the autocorrelation of $\mathbf{u}$ from Part II: the spike and its sidelobes.

Decoding returns a multiple of $\mathbf{v}$ exactly when $|\mathcal{F}\mathbf{u}|^2$ is constant: a flat spectrum. **Unitary** vectors are the ones with every $|(\mathcal{F}\mathbf{u})_k| = 1$, so the multiple is 1. For $\mathbf{r} = \frac{1}{2}(1, 1, 1, -1)$, $\mathcal{F}\mathbf{r} = (1,\; -i,\; 1,\; i)$. For the shift $\boldsymbol{\delta}_1 = (0, 1, 0, 0)$, $\mathcal{F}\boldsymbol{\delta}_1 = (1,\; -i,\; -1,\; i)$: the shift factors of Step 2, all of length 1.

In practice HRR is computed this way: two FFTs, one coordinate-wise product, one inverse FFT.

The Exact Inverse Divides

The exact inverse has spectrum $1/\mathcal{F}\mathbf{c}$: it divides each frequency by the atom's coefficient there. Take $\mathbf{c} = (1,\ 1,\ 1,\ 0.9)$:

$$|\mathcal{F}\mathbf{c}| = (3.9,\ 0.1,\ 0.1,\ 0.1) \quad \Longrightarrow \quad 1/|\mathcal{F}\mathbf{c}| = (0.26,\ 10,\ 10,\ 10)$$

and transforming back gives $\mathbf{c}^{-1} = (2.56,\ -7.44,\ 2.56,\ 2.56)$, more than four times as long as $\mathbf{c}$.

	exact inverse $\mathbf{c}^{-1}$	**involution $\mathbf{c}^{*}$**
multiplies frequency k by	$1/\mathcal{F}_k\mathbf{c}$	$\overline{\mathcal{F}_k\mathbf{c}}$
on this $\mathbf{c}$, frequencies 1–3	multiplied by 10	multiplied by 0.1
on a trace with one pair	exact	distorted where $\mathbf{c}$ is weak
on a trace with several pairs	amplifies the other pairs where $\mathbf{c}$ is weak	no division, so no blow-up

That is what we measured at $n = 1{,}024$: with one pair, 1.00 for the exact inverse against 0.71 for the involution; with two pairs, 0.34 against 0.58. There a random atom's smallest coefficient is typically about 0.04, a gain of about 28 on the other pairs. The involution trades exactness on clean traces for safety on crowded ones; unitary atoms need no trade.

Fourier Transforms Inside NLP Systems

system	where the Fourier transform enters
HRR; HoIE (Nickel et al. 2016)	binding, or scoring a knowledge-graph triple, by circular convolution or correlation, computed with FFTs in $O(n \log n)$
sinusoidal position encodings (Vaswani et al. 2017)	each pair of dimensions is a wave over word positions, at its own frequency; moving k positions turns each pair by an angle set by k and that pair's frequency
RoPE (Su et al. 2024)	each pair of query or key coordinates, read as a complex number, is multiplied by $e^{im\theta}$ at position m, so a score depends on positions only through their offset: in Fourier coordinates, binding with a unitary vector
FNet (Lee-Thorp et al. 2022)	each self-attention sublayer replaced by a Fourier transform with no parameters: 92–97% of BERT's accuracy on GLUE, training 80% faster on GPUs at 512 tokens

What the rows share is Step 2's waves, over the coordinates of a vector (HRR, HoIE) or over the positions in a sentence (the rest).

Part VI
Where HRR Came From

Why "Holographic," and Where HRR Came From

1960s–70s	holographic and correlation memories: store a pair by convolving, recall by correlating, by analogy with optical holograms
1982	psychological models of memory for pairs built on convolution and correlation: Murdock's TODAM and Metcalfe Eich's CHARM
1990	Hinton's *reduced descriptions*: a whole structure compressed into a vector that fits one slot; Smolensky's tensor product binding
1991–95	Plate's HRR: circular convolution at fixed n, the involution, cleanup memory, a capacity analysis
2003	Plate's book, *Holographic Reduced Representation*

"Holographic" because convolution and correlation are the operations of those holographic memories, and, as in an optical hologram, every item is spread over every coordinate: damage to some coordinates lowers every cosine a little rather than erasing any one item. "Reduced" because a whole frame becomes one vector that can fill a slot of another frame, as in *Hunger caused Mark to eat the fish*.

So convolution memory was a model of human recall before Plate made it a representation for structure.

HRR Among Its Neighbors

	TPR (Smolensky)	**HRR** (Plate)	**MAP** (Gayler)
atoms	any vectors	random, variance $1/n$	random ± 1
binding	outer product $\mathbf{r}\mathbf{f}^\top$	circular convolution	coordinate-wise product
size of a pair	n^2	n	n
unbinding	exact, with orthonormal roles	approximate; exact for unitary roles	exact: binding is its own inverse
commutative?	no	yes	yes
cost per pair	n^2	$n \log n$, by FFT	n

Next session: MAP. With coordinates ± 1, multiplying a vector by itself coordinate-wise gives all ones, so every atom undoes itself exactly. Unbinding becomes exact and computable by hand, with no Fourier analysis.

That frees the next session for the architecture as a whole: the three operations, cleanup, records, sequences, analogy without training, and types. Everything there has an HRR counterpart from today.

Summary

1. HRR atoms are random vectors with variance $1/n$: length about 1, and nearly orthogonal to each other.
2. **Binding** is circular convolution: n numbers, similar to neither input, similarity-preserving, and commutative. Order is carried by which role a filler is bound to.
3. **Unbinding** is correlation with the role, $\mathbf{c} \# \mathbf{t} = \mathbf{c}^* \circledast \mathbf{t}$, where the involution keeps the first coordinate and reverses the rest.
4. Decoding returns the filler plus the role's **sidelobes** convolved with it. At large n the sidelobes are small but many, so one pair decodes at about 0.71; unitary roles have none and decode exactly.
5. With k pairs the right answer scores about $1/\sqrt{k+1}$, against a best wrong answer near $\sqrt{2 \ln m}/\sqrt{n}$. Capacity grows linearly in n and fails as a cliff.
6. **Cleanup** rounds to the stored atoms and frames. Nested structures need it at every level.
7. Waves are the vectors a shift only rescales, so convolution multiplies each wave by one number, its **Fourier coefficient**: $\mathcal{F}(\mathbf{u} \circledast \mathbf{v}) = \mathcal{F}\mathbf{u} \odot \mathcal{F}\mathbf{v}$. The FFT computes all n coefficients in about $(n/2) \log_2 n$ steps, so a binding costs $O(n \log n)$ rather than n^2.
8. Decoding multiplies coefficient k by $|(\mathcal{F}\mathbf{c})_k|^2$, so unitary roles decode exactly. The exact inverse divides by the coefficients and amplifies the other pairs, so Plate decoded with the involution.

Exercises

1. By hand at $n = 4$: superpose the two pairs $\mathbf{r} \circledast \mathbf{v}$ and $\mathbf{u} \circledast \mathbf{v}$ into one trace, with $\mathbf{r} = \frac{1}{2}(1, 1, 1, -1)$, and decode with $\mathbf{r}$. Show that $\mathbf{r}$'s sidelobes contribute nothing, and compute what the other pair contributes.
2. By hand at $n = 4$: compute $\mathcal{F}\mathbf{v}^*$ for $\mathbf{v}^* = (2, 1, 1, 0)$ and check that it is the conjugate of $\mathcal{F}\mathbf{v}$. Then run the FFT diagram on $\boldsymbol{\delta}_1 = (0, 1, 0, 0)$ and say why every coefficient has length 1.
3. In numpy, implement $\circledast$ and $\#$ with `np.fft`, using `np.roll(c[::-1], 1)` for the involution. Reproduce the 0.71 of Part II at $n = 1{,}024$.
4. Build *Mark ate the fish*, *The fish ate Mark*, and *John saw Mark*. Reproduce the similarity table of Part III, then make john and mark similar atoms ($\cos = 0.6$) and report what changes.
5. Reproduce the capacity table for $n = 256$ and $n = 1{,}024$, and plot accuracy against k/n. Where is the cliff, as a fraction of n?
6. Make every role unitary by setting each Fourier coefficient's magnitude to 1. Measure the gain for $k = 1, 3, 10$, and compare with $1/\sqrt{k}$.
7. $\star$ Encode *Hunger caused Mark to eat the fish*. Query *Who ate?* with and without cleanup of the intermediate frame, then add a third level of nesting and report the cosine at each depth.

Reading

- T. A. Plate, Holographic reduced representations, *IEEE Transactions on Neural Networks* 6(3), 1995. The source for today: atoms, the involution, frames, capacity.
- T. A. Plate, *Holographic Reduced Representation: Distributed Representation for Cognitive Structures*, CSLI Publications, 2003.
- G. E. Hinton, Mapping part-whole hierarchies into connectionist networks, *Artificial Intelligence* 46, 1990. Reduced descriptions.
- P. Smolensky, Tensor product variable binding and the representation of symbolic structures in connectionist systems, *Artificial Intelligence* 46, 1990.
- B. B. Murdock, A theory for the storage and retrieval of item and associative information, *Psychological Review* 89, 1982.
- J. Metcalfe Eich, A composite holographic associative recall model, *Psychological Review* 89, 1982.
- J. W. Cooley & J. W. Tukey, An algorithm for the machine calculation of complex Fourier series, *Mathematics of Computation* 19, 1965. The FFT.
- P. Kanerva, Hyperdimensional computing, *Cognitive Computation* 1, 2009. The wider family, for next session.